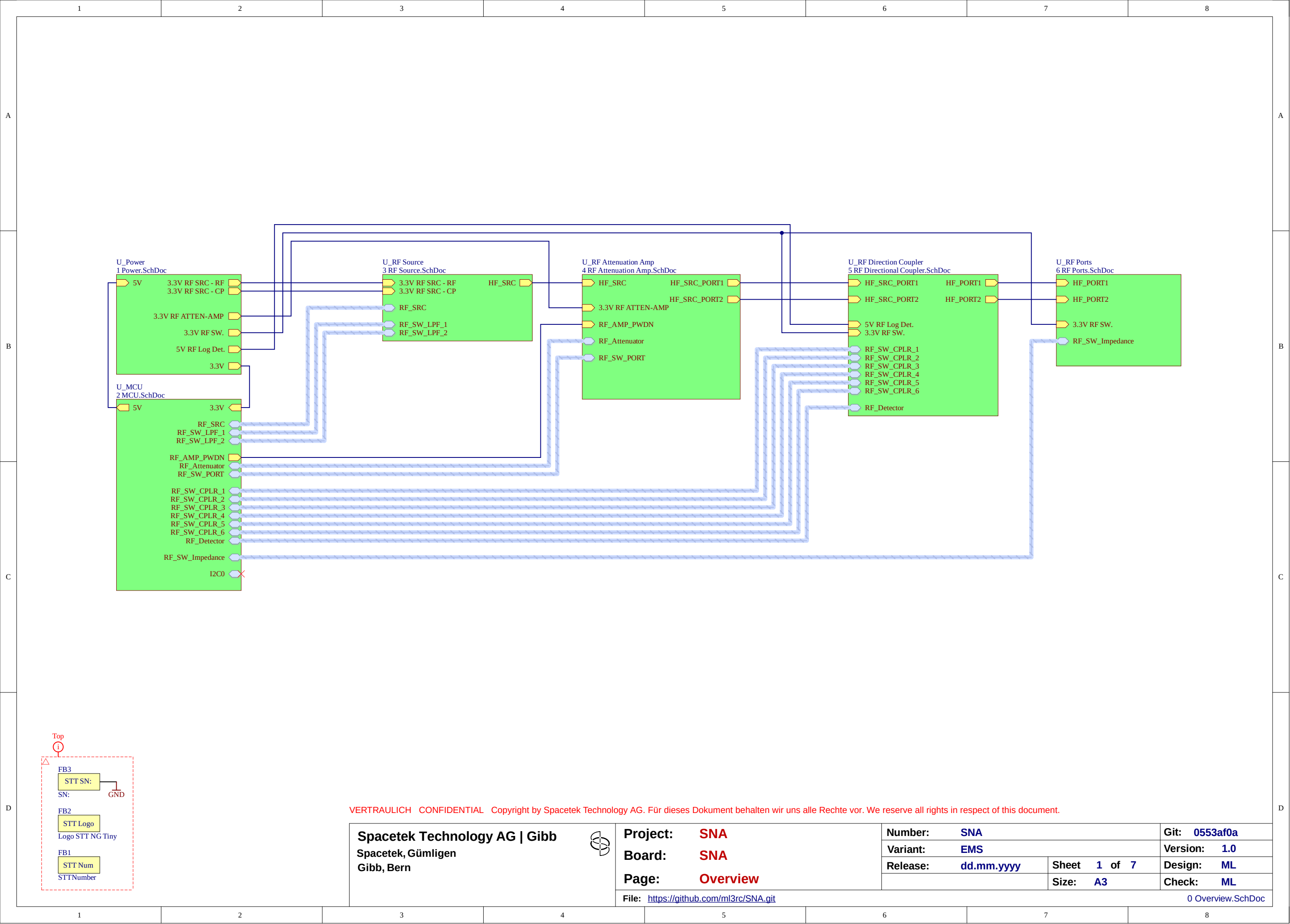
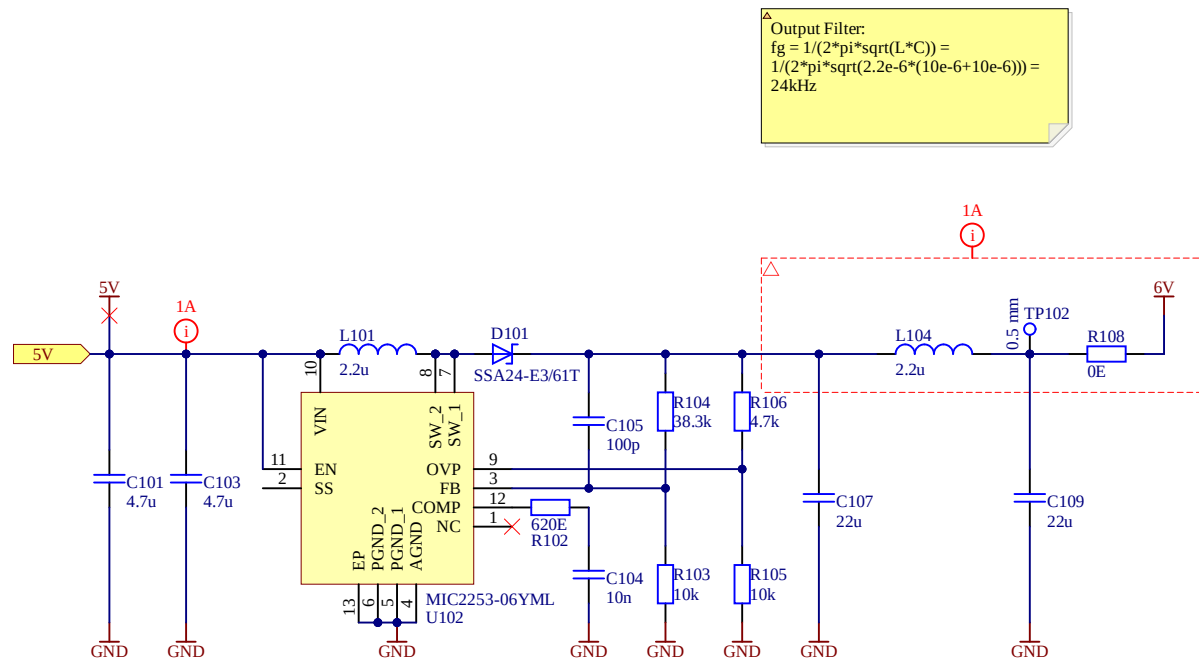
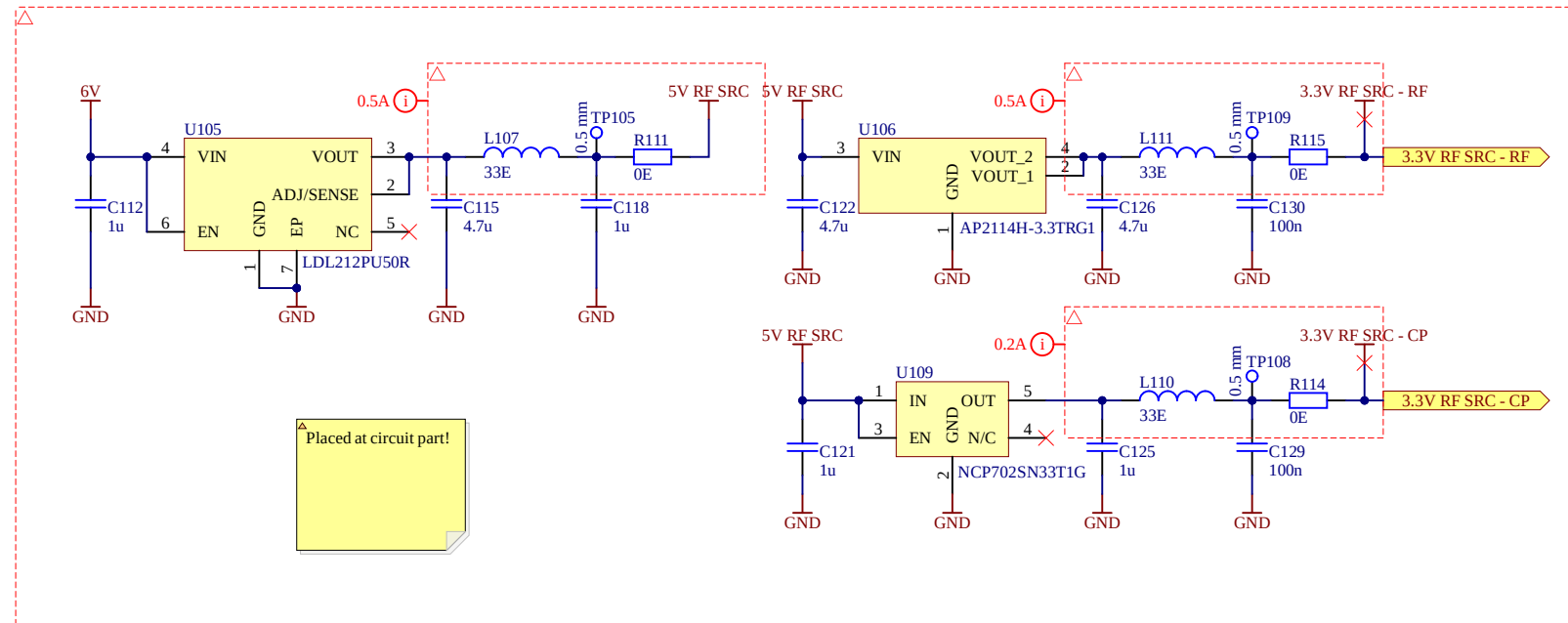






Output Filter:

$$f_g = 1/(2\pi \sqrt{L \cdot C}) = 1/(2\pi \sqrt{2.2 \times 10^{-6} (10^{-6} + 10^{-6})}) = 24 \text{ kHz}$$



Placed at circuit part!

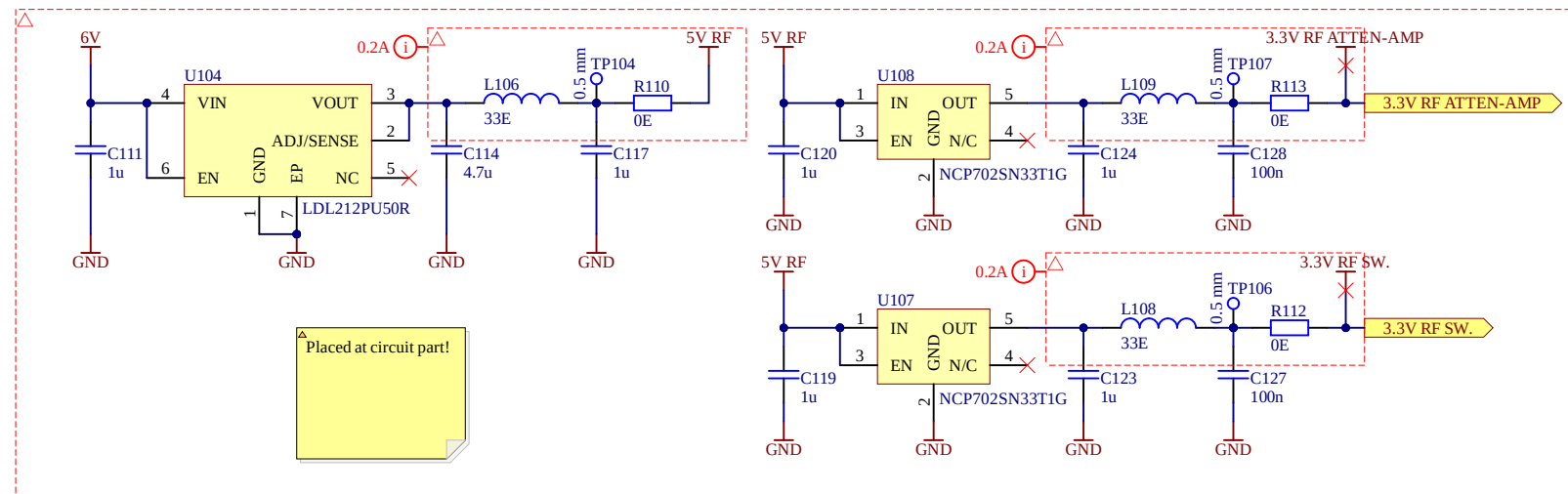
$L = 2.2\mu\text{H}$ as per reference design
 $C_{out} = 22\mu$ as per reference design

Set ss to 10ms
 $C_{ss} = (T_{ss} - 1\text{ms}) / 85k\Omega = (10\text{ms} - 1\text{ms}) / 85k\Omega = 105\text{nF} \rightarrow 100\text{nF}$ (10nF internal cap!)

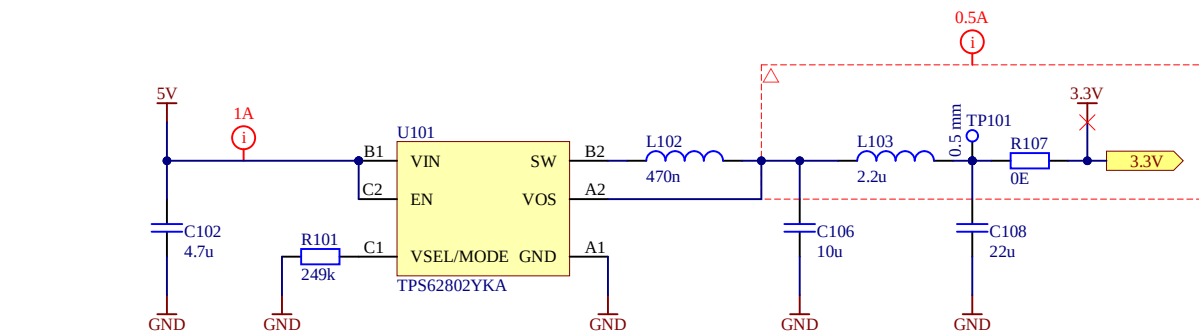
 $U_{ovp} = 8\text{V}$
 $R_{bottom} = 10k$
 $R_{top} = (U_{ovp} * 15k\Omega * R_{bottom}) / (1.245 * 67k\Omega) - R_{bottom} = (8\text{V} * 15k\Omega * 10k\Omega) / (1.245 * 67k\Omega) - 10k\Omega = 4.39k\Omega \rightarrow 4.7k\Omega$

 $U_{out} = 6\text{V}$
 $U_{fb} = 1.245\text{V}$
 $I_{fb} = -450\text{nA}$
 $R_{bottom} = 10k\Omega$
 $R_{top} = (U_{out} - U_{fb}) / (U_{fb} / R_{bottom} + I_{fb}) = (6\text{V} - 1.245\text{V}) / (1.245\text{V} / 10k\Omega - 450\text{nA}) = 38.3k\Omega$

 $C_{comp} = 10\text{nF}$ as per reference design
 $R_{comp} = 620\Omega$ as per reference design



Placed at circuit part!



Output Filter:

$$f_g = \frac{1}{2\pi \sqrt{L \cdot C}} = \frac{1}{2\pi \sqrt{2.2 \times 10^{-6} (10 \times 10^{-6} + 10 \times 10^{-6})}} = 24 \text{ kHz}$$

Design from WEBENCH

For current specs see power blockdiagramm
 20mil -> 1A (2.14A)
 15mil -> 0.5A (1.79A)
 12mil -> 0.2A (1.56A)

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Project:	SNA
Board:	SNA
Page:	Power

File: <https://github.com/ml3rc/SNA.git>

Number: SNA

Variant: EMS

Release: dd.mm.yyyy

Sheet 2 of 7

[illegible]

Size:	A3
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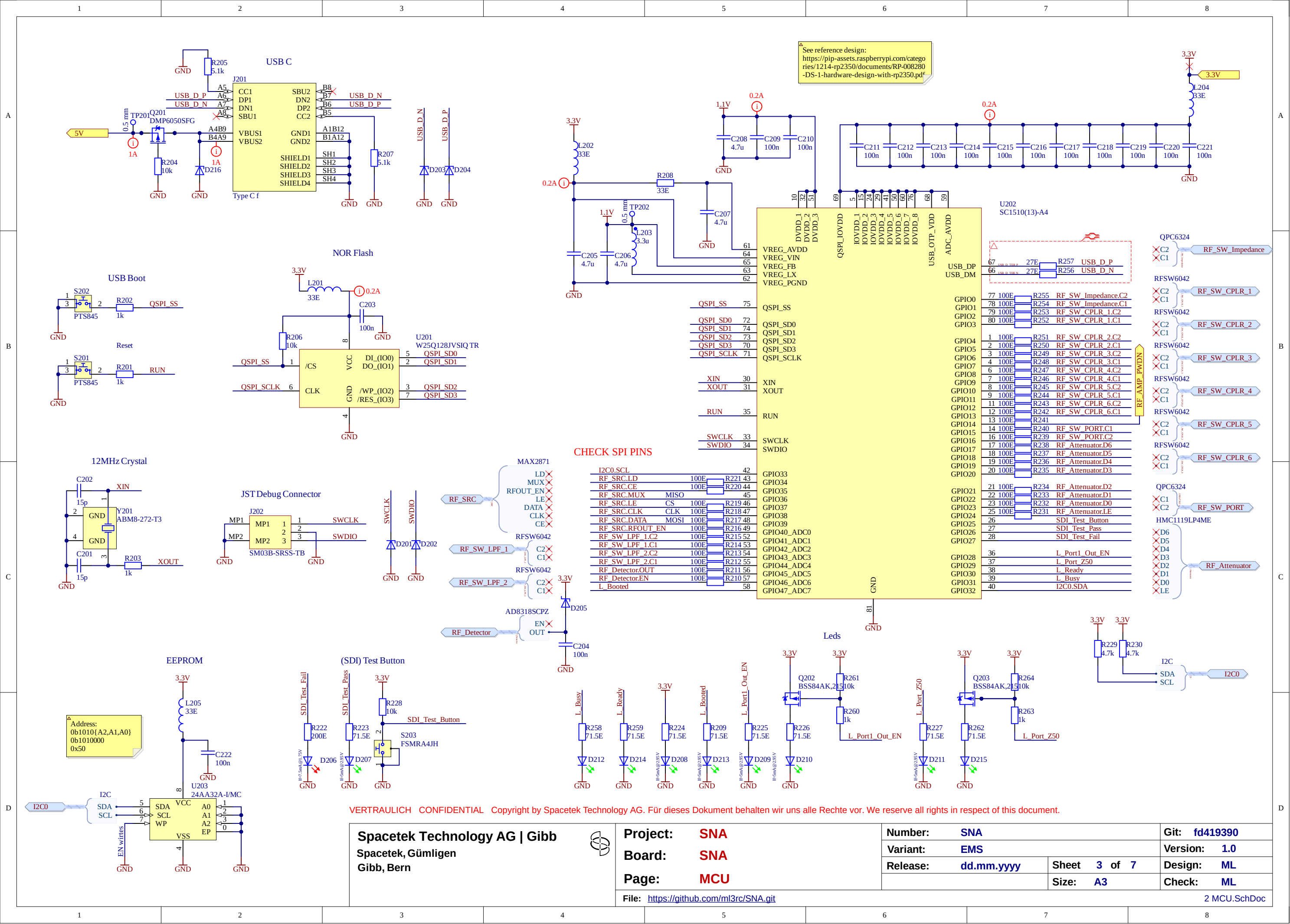
Git: 0553af0a

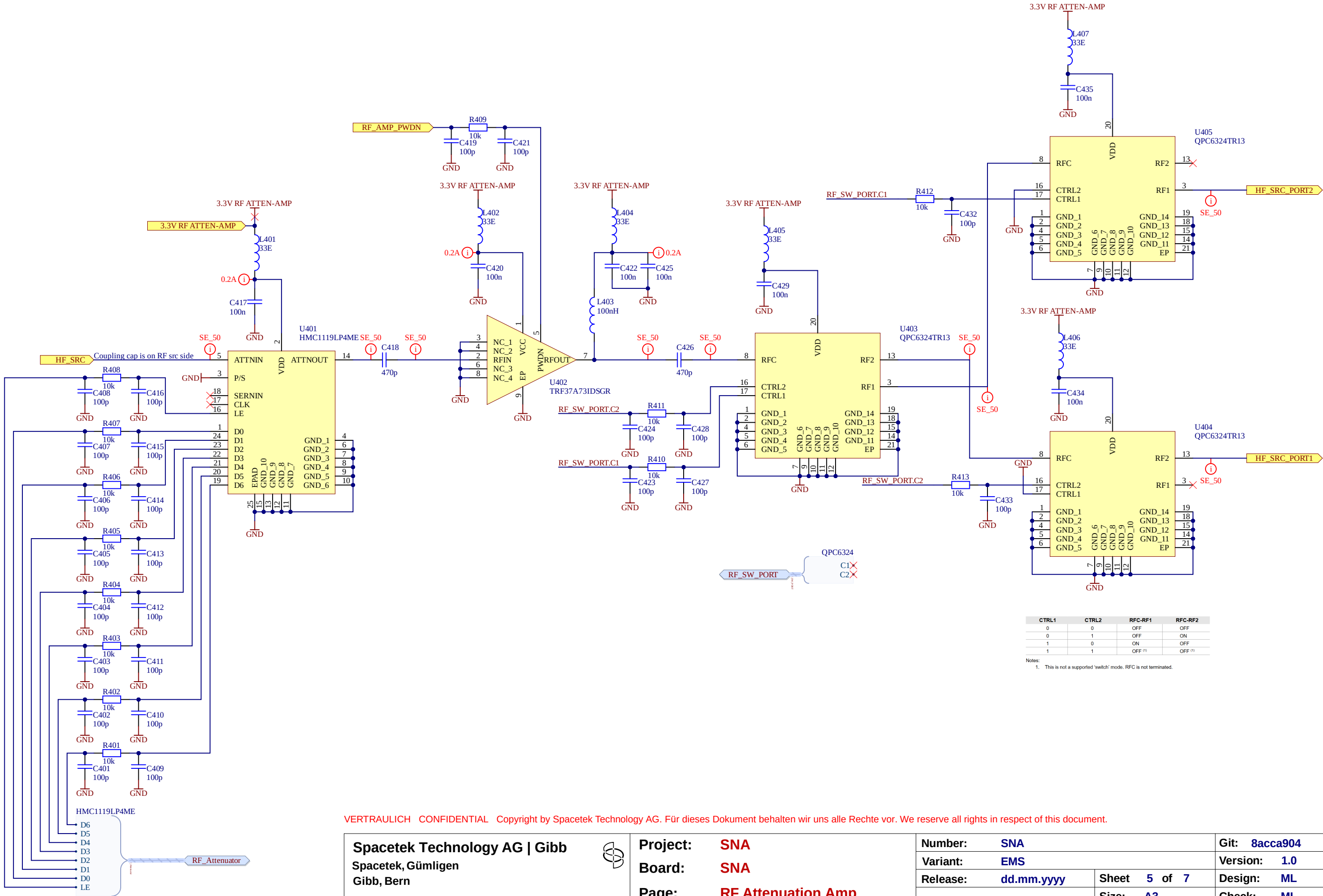
Version:	1.0
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Design:	ML
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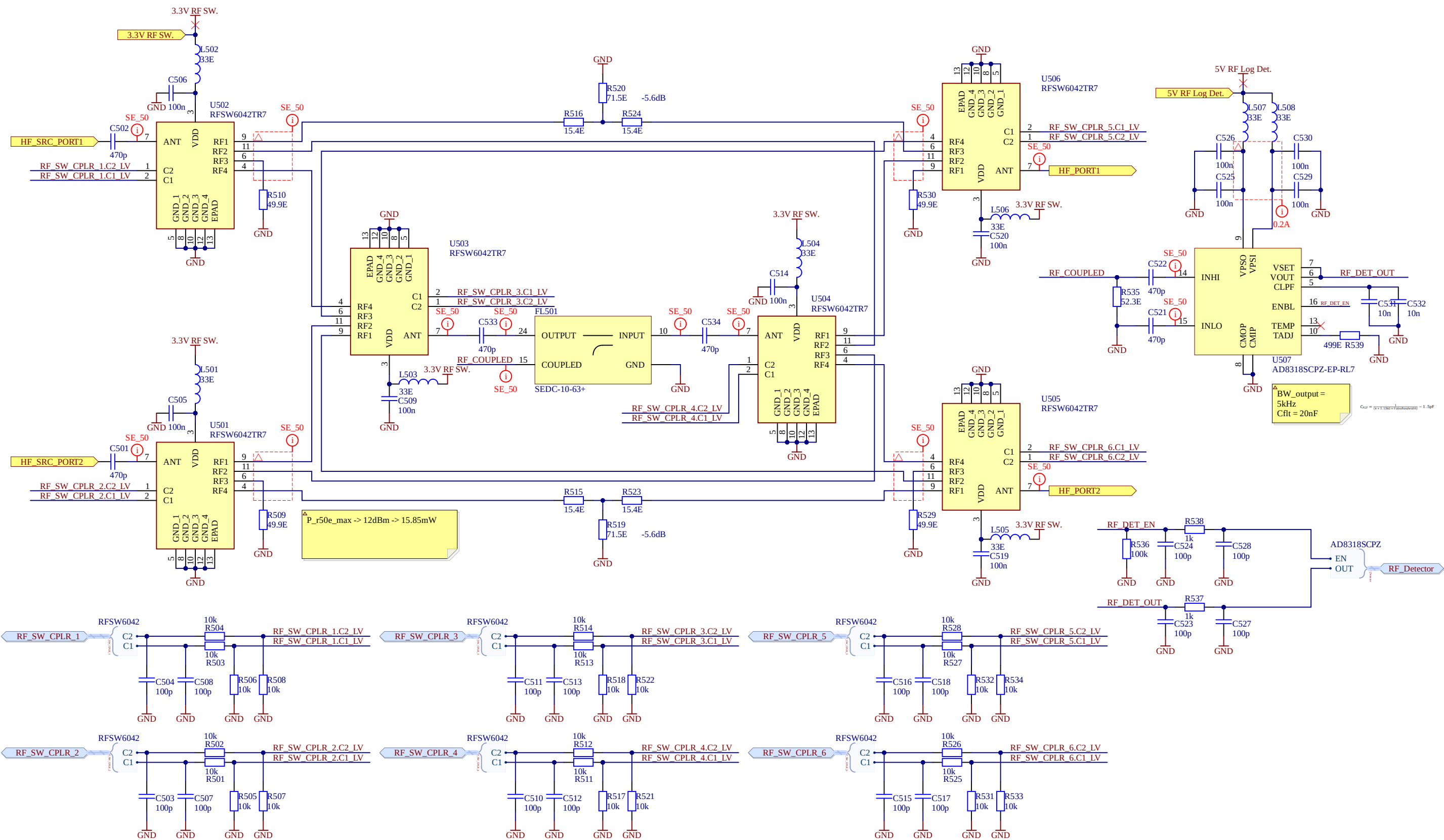
Check:	ML
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1 Power.SchDoc

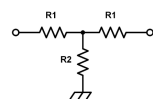




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△ T-Pad calculated with:
<https://www.allaboutcircuits.com/tools/t-pad-attenuator-calculator/>



Inputs

Attenuation (dB)

Impedance (Ω)

Calculate

Outputs

R_1 (Ω)

R_2 (Ω)

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Project: SNA
Board: SNA
Page: RF Directional Coupler

File: <https://github.com/ml3rc/SNA.git>

Number: SNA		Git: fd419390
Variant: EMS		Version: 1.0
Release: dd.mm.yyyy	Sheet 6 of 7	Design: ML
	Size: A3	Check: ML

5 RF Directional Coupler.SchDoc

50-75 Matching Pad:
 $\text{solve}(\{75 = ((1)/(((1)/(R_p)) + ((1)/(50)))) + R_s, 50 = ((1)/(((1)/(75 + R_s)) + ((1)/(R_p))))\})$
 $R_p = 86.6\Omega$
 $R_s = 43.3\Omega$
Insertion Loss = 5.72dB

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Project: **SNA**
Board: **SNA**
Page: **RF Ports**

File: <https://github.com/ml3rc/SNA.git>

Number: **SNA**

Variant: **EMS**

Release: **dd.mm.yyyy**

Sheet **7** of **7**

Size: **A3**

Git: **0553af0a**

Version: **1.0**

Design: **ML**

Check: **ML**

6 RF Ports.SchDoc